Same Score, Different Evidence: Decodability, Surface Sufficiency, and Causal Relevance in Code Models

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Abstract

Interpretability experiments must distinguish model mechanisms from measurement artifacts. We present a case-study-backed reporting matrix for linear probes over five identifier properties and roles in code models. The study includes three models and collectively covers all seven XLCoST programming languages without forming a complete factorial design. The matrix binds each claim to an estimand, comparator, matching unit, uncertainty statement, outcome, and falsifier. Two matched cases motivate it. On Python, boolean occurrence type is decodable at 0.981–0.988 macro-F1, while a masked source-line classifier reaches 0.983 without a language model. Paired problem-clustered deltas are -0.003 to $+0.002$, with intervals excluding probe advantages above 0.014–0.017, although the sign of that difference varies by language and by comparator. A full-residual patch at a class or structure query-name site also fails its specified bidirectional gate in Qwen2.5-1.5B, recovering 0.009 to 0.020 of the matched behavioral gap. This bounds the effect at that site rather than showing global non-use. A third case exposes a failed control. Role-conditioned iterator renaming yields 0.998–0.999 F1 at embedding layer zero, consistent with a lexical code introduced by the intervention rather than semantic robustness. Decodability, capacity control, cue sensitivity, surface sufficiency, and site-state causal relevance therefore require separate, outcome-aware records.

1 Introduction

Linear probes are often used to ask what information a representation contains. A successful probe establishes decodability under a particular readout. It does not by itself establish that the model computed an abstract feature, that the feature is unavailable in surface form, or that the model uses it behaviorally [Belinkov, 2022, Voita and Titov, 2020]. Control tasks constrain probe capacity [Hewitt and Liang, 2019], and targeted studies show that probe accuracy need not entail task relevance [Ravichander et al., 2021, Elazar et al., 2021]. Capacity is only one source of ambiguity. Input leakage, label construction, sample resolution, and an unmatched intervention can each preserve high accuracy while changing the scientific claim.

This distinction is central to interpretability as a science. A measurement is useful when its target is explicit, alternative explanations are tested, and contrary outcomes change the conclusion. We use identifier probing as a comparative case study. Some targets are semantic variable roles, such as accumulating values or indexing a container [Sajaniemi, 2002]. Others are occurrence types or declared type identifiers. Names, syntax, and context can all predict these targets, which makes probe accuracy underidentified. Prior code-probing studies, including the 15-task INSPECT framework, map decodable syntactic and semantic information across layers and models [Karmakar and Robbes, 2021, Troshin and Chirkova, 2022, Karmakar and Robbes, 2024]. We do not propose another probe,

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Table 1: Target coverage. No row should be read as a complete three-model by seven-language experiment. Table 3 gives the outcome-aware evidence records.

Target	Models	Language coverage
Accumulator	Qwen2.5-1.5B	Python plus Java, C++, C, C#
Index or key	Qwen2.5-1.5B	Python, Java, JavaScript, C++, C, C#
Iterator	All three	Python, Java, JavaScript, PHP, C++, C, C#
Boolean	All three	Python, Java, JavaScript, PHP
Class or structure	All three	Python, C++, JavaScript, C

control, or task taxonomy. Existing work catalogs probing limitations and shows that activation-patching conclusions depend on metric and intervention choices [Belinkov, 2022, Zhang and Nanda, 2023, Heimersheim and Nanda, 2024]. Subspace interventions can also create an illusion of localization [Makelov et al., 2023], which motivates calls for falsifiable interpretability designs [Leavitt and Morcos, 2020] and for naming the causal mediator a claim is about [Mueller et al., 2024]. Reporting standards in other empirical sciences answer a related need by binding a published claim to its estimand, comparator, and uncertainty statement [Schulz et al., 2010]. Our distinct contribution is an outcome-aware reporting unit that binds each claim to its estimand, comparator, matching unit, uncertainty, observed outcome, and falsifying next test. Unlike a control checklist, the record preserves failed and confounded comparisons and states how they change the licensed claim.

Our evidence covers five roles, three models, and the seven-language XLCoST corpus [Zhu et al., 2022]. The experiment is intentionally reported as an uneven evidence matrix rather than a full factorial study. Qwen2.5-1.5B, Qwen2.5-Coder-1.5B, and StarCoder2-7B all appear in the controlled boolean, class or structure, and iterator experiments [Qwen et al., 2024, Hui et al., 2024, Lozhkov et al., 2024]. Earlier accumulator and index runs use one model and a weaker split protocol. Iterator runs cover all three models and all seven languages. This is a methodological case study of evidential boundaries, not a validation of the matrix as a universal framework. Its heterogeneity separates replicated findings from leads that still require confirmation.

The central result is that high probe F1 produces different evidential outcomes. The paired boolean comparison supports a bounded negative. The class or structure intervention fails its gate at one site and layer. Role-conditioned iterator renaming produces near-perfect embedding-layer scores, revealing that a nominal control can introduce a new shortcut. Legacy accumulator and index effects remain motivating observations. A single accuracy number, or a checkmark indicating that a control ran, cannot identify among these explanations.

2 Study design and evidence matrix

Targets, models, and languages. We treat five identifier properties and roles as related case studies, not measurement-equivalent instances of one construct. Accumulators update within loops. Indices or keys appear as counters or subscripts. Iterators are loop-bound names and their uses. Boolean labels distinguish assignment, conditional, loop, and return occurrences. Class or structure labels mark declared type names and references. Binary token probes label all other tokens as negative. Language-specific AST and regular-expression extractors lack a common multilingual gold audit. Protocols also differ. Accumulator and index use legacy token splits and test-maximized layers. Boolean mean-pools occurrences and uses problem groups. Iterator and class or structure use token labels, problem groups, validation-selected layers, and five seeds. XLCoST contains C, C++, C#, Java, JavaScript, PHP, and Python. Table 1 states coverage. Scores across rows do not estimate one common task.

Four comparison questions. Renaming asks whether lexical identity carries the prediction. Random-noun, single-character, and numeric strategies are role-agnostic textual substitutions. The misleading strategy is not role-agnostic. Target identifiers receive counter-role names while other identifiers receive role-looking names. The vocabulary therefore encodes the label. The all-same strategy is also non-bijective. Because every condition refits its probe and layer, neither strategy tests semantic robustness or a fixed invariant direction. We retain them as diagnostics of a failed intervention. A model-free character classifier with the target name masked asks whether local

Table 2: Case outcomes after interpreting the available comparisons. Values are macro-F1. A diagnostic can reveal a failed control without supporting a representation claim.

Case	Probe	Comparison outcome	Strongest supported inference
Legacy accum./index	0.893, 0.959	$\Delta = +0.079, -0.272$	Exploratory cue response under a weaker protocol
Iterator	0.976–0.987	role-conditioned rename reaches 0.998–0.999 at L0	The intervention encodes a lexical shortcut
Boolean	0.981–0.988	paired $\Delta \in [-0.003, +0.002]$	Probe advantage over masked local source excluded above 0.017
Class/site	0.979–0.982	patch recovery 0.009, 0.020	Bounded site-state effect at the tested site and layer

source text already determines the label. Randomized-label controls ask whether the probe family can fit a specified null. Boolean assigns a random label per variable name. Iterator and class or structure permute labels within programs. Full-residual patching asks whether replacing the complete site state moves a behavioral logit readout [Meng et al., 2022]. Its interpretation depends on the corruption, metric, and readout, so these choices belong in the evidence record [Zhang and Nanda, 2023, Heimersheim and Nanda, 2024].

These comparisons are not interchangeable. Role-agnostic refit renaming can measure recoverability after a cue changes, but role-conditioned renaming can create a cue. Positive selectivity can reject its specified permutation task while leaving a surface shortcut intact. A model-free comparator can establish surface sufficiency but cannot show how the model uses the feature. Full-residual patching tests site-state relevance, not use of the decoded probe direction, and only at its specified layer, span, direction, and readout.

Evidence grades. Boolean, iterator, and class or structure experiments use problem groups, validation-selected layers, five seeds, and BCa or clustered uncertainty. Iterator additionally reports a within-program label-permutation control and a name-only comparator, but their difference is not a paired clustered interval. Accumulator and index use token splits, test-maximized layers, and whole-word renaming. We retain their opposite effects as legacy hypotheses, not population estimates. A DeepSeek index run is excluded because a tokenizer offset error invalidates its activation alignment.

3 The same probe score supports different claims

3.1 Renaming diagnoses an intervention confound

Legacy Qwen2.5-1.5B runs first motivated the comparison. The index probe falls from 0.959 to 0.687 under misleading names, a change of -0.272, while the accumulator probe rises from 0.893 to 0.972, a change of $+0.079$. Token-level splitting, test-maximized layers, role-conditioned replacements, and refitting prevent a robustness interpretation. These are exploratory cue responses rather than a confirmed cross-role dissociation.

The modern iterator rerun reveals the intervention problem directly. Across the three models, baseline F1 is 0.976, 0.978, and 0.987. The widest BCa 95% interval is $[0.964, 0.982]$, and permutation selectivity is $+0.526$ to $+0.552$. Role-conditioned misleading names change F1 by $+0.023$, $+0.020$, and $+0.012$, but all three probes select embedding layer zero and reach 0.998–0.999. The name pools encode target membership, so this result is compatible with relearning an inverted lexical rule. It does not support semantic robustness. Single-character and numeric names reduce F1 by 0.046–0.072. A name-only classifier on the same 427 Python programs reaches 0.918, leaving an unpaired gap of 0.058–0.069. Python-trained transfer ranges are 0.934–0.981, 0.898–0.923, and 0.887–0.938. Because target evaluation is not separated by one global cross-language problem partition, these scores establish descriptive transfer rather than language abstraction. Exploratory iterator patching appears in the appendix and is not the class or structure causal gate.

3.2 A model-free baseline bounds the boolean result

Boolean occurrence type provides the clearest bounded negative result. On a frozen Python sample, the three models reach 0.982, 0.981, and 0.988 macro-F1. Selectivity ranges from $+0.272$ to $+0.288$. These values reject the specific hypothesis that an equally expressive probe predicts labels randomly assigned per variable name as well as it predicts the real usage labels. Yet a character classifier on the enclosing source line, with the variable name masked, reaches 0.983. The probe-minus-surface differences are -0.001, -0.002, and $+0.005$.

The original exports retained no aligned predictions, so we reran both predictors on the frozen sample and used aligned occurrence identifiers. Pooling the five seed test folds gives $2{,}067$ seed-level predictions for $1{,}410$ unique occurrences in 915 of the $1{,}301$ source problems, and predictor overlap is complete on every fold. The committed artifact retains the aggregate paired summary, not those prediction-level inputs. The probe-minus-surface estimates and problem-clustered 95% intervals are -0.002 $[-0.017, 0.014]$, -0.003 $[-0.037, 0.014]$, and $+0.002$ $[-0.016, 0.017]$. Every interval covers zero and excludes a probe advantage larger than 0.014–0.017 macro-F1. The masked-line comparator was the best of the reported surface feature families on this frozen sample, so the bound is conditional on that non-nested selection. That selection matters. Against the next-best Python comparator, the masked enclosing statement at 0.970, the differences are $+0.010$ to $+0.017$. Across the four committed languages, probe minus best baseline ranges from -0.004 to $+0.021$, the largest value being PHP with StarCoder2-7B, and line masking is degenerate outside Python because detokenized XLCoST programs there occupy a single line. Five refit renaming conditions change F1 by at most 0.013, but the role-conditioned and non-bijective strategies prevent a robustness claim. The supported conclusion is narrower. On Python, the probe does not outperform the selected masked local-syntax comparator by more than 0.014–0.017. The other languages have unpaired differences only, so they carry no bound.

3.3 A site-state intervention fails its specified gate

The class or structure experiments and the boolean experiments use different controls. Across all three models, baseline F1 is 0.979–0.982 and within-program permutation selectivity is $+0.503$–$+0.553$. Misleading class-style names change F1 by -0.004, -0.004, and $+0.004$, but this role-conditioned vocabulary prevents a robustness interpretation. Python-trained probes transfer to C++, JavaScript, and C at 0.885–0.980. These results support decodability and performance above the specified within-program permutation task. They do not establish abstraction or a fixed rename-invariant direction.

We then test site-state relevance in Qwen2.5-1.5B by patching the full residual state. The experiment contains 288 matched class-versus-function prompts in 48 template clusters. Let $D = \ell(\texttt{True}) - \ell(\texttt{False})$. Denoising patches the class-source residual into a function prompt and defines $e_d = D_{\text{patched function}} - D_{\text{function}}$. Noising reverses the patch and defines $e_n = D_{\text{class}} - D_{\text{patched class}}$. Recovery is the ratio of the mean signed effect to the mean matched gap $D_{\text{class}} - D_{\text{function}}$. Percentile intervals resample template clusters. The gate requires positive, control-separated effects, recovery of at least 0.05, and mean effects of at least 0.10 in both directions. These thresholds are design choices rather than calibrated behavioral minima.

Both prompt classes prefer `True`, which limits the behavioral readout's effective decision range. Their mean logit differences are $+2.75$ for class prompts and $+1.78$ for function prompts, although the matched class-minus-function gap is $+0.97$ and has the expected sign for 98.6% of pairs. The gate therefore tests whether patching transfers this relative distinction, not whether it flips a well-calibrated binary decision.

Denoising produces a mean effect of 0.0087 with a 95% interval $[-0.0007, 0.0188]$ and recovery 0.0089. Noising produces 0.0199 $[0.0109, 0.0296]$ with recovery 0.0204. The full-residual intervention fails the specified gate. The protocol and results entered the repository together, so we do not claim auditable preregistration. Recovery divides by the matched gap, so a compressed decision range does not explain a ratio of 0.009 to 0.020. We read this as a bounded negative at the tested site, layer, and span rather than evidence that class information is absent or unused. The intervals exclude mean effects above 0.0188 and 0.0296 logit-difference units in the tested directions. Decodability and selectivity therefore do not establish a substantive site-state contribution under this readout.

Table 3: Outcome-aware evidence records. "Unpaired" means that uncertainty on the difference is unavailable. Each next test can change the stated claim.

Case	Comparison and outcome	Matching and uncertainty	Claim and falsifier
Legacy roles	Refit accumulator $+0.079$, index -0.272	Token split, test-selected layer, no interval	Exploratory only. Rerun grouped with label-blind renames
Iterator	Role-conditioned rename reaches L0 at 0.998–0.999	Problem split and marginal BCa intervals	Control is confounded. Use bijective role-agnostic renames
Boolean	Probe minus masked line is -0.003 to $+0.002$	915 clusters with paired 95% intervals	Not supported on Python. No selected-baseline advantage above 0.017. Nest selection and add languages
Class/site	Bidirectional residual-patch gate fails	48 template clusters with 95% intervals	Bounded effect at one site and layer. Test other spans and layers

4 An outcome-aware reporting matrix

The contribution is a reporting obligation, not a new control or evidence ladder. Each record states the target and prediction unit, claim and estimand, comparator and matching rule, uncertainty, outcome, and falsifying next test. We distinguish five claim types. Decodability exceeds a label baseline. Capacity control exceeds a specified randomized-label task. Cue sensitivity uses an intervention that does not itself encode the label. Surface sufficiency compares a model-free predictor on a matched population. Site-state causal relevance requires a gate-specified intervention and behavioral readout. No claim entails another.

These heterogeneous cases demonstrate how the record changes interpretation. They do not validate the matrix as universal. A claim of cross-language abstraction remains open until one extractor, global problem partition, and matched control battery cover a complete language grid. Replication packages should publish the record beside metric tables and state the smallest effect the design can exclude.

5 Conclusion

High probe accuracy is not a common unit of interpretability evidence. In these cases, matched surface prediction supports a bounded negative, role-conditioned renaming exposes a shortcut, and a specified patching gate fails at one site and layer. Publishing each claim as an outcome-aware record lets null and confounded outcomes constrain conclusions rather than disappear behind a control checkmark.

6 Limitations

The study collectively covers three models, five roles, and seven languages, but not a complete $3 \times 5 \times 7$ design, and Table 1 gives the per-target coverage. The targets are heterogeneous, and their language-specific extractors lack a common gold audit. Role-conditioned and all-same renames can encode or alter labels. Legacy accumulator-index results lack problem groups and scope-aware renaming. Iterator has no paired probe-minus-surface interval, and its target evaluation does not enforce one global cross-language problem partition. The boolean bound is conditional on a surface comparator selected on the frozen sample, its sign varies across languages, and its prediction-level paired inputs are not retained. Full-residual patching covers one model, layer, site, and readout. The gate and stopping rule were not registered in advance. These cases motivate the reporting matrix rather than validate it.

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A Per-role control results

Table 4: Complete boolean occurrence-type probe and model-free baseline results. The best baseline is selected from name-only, masked statement, masked line, masked window, covariate-only, and majority features. ρ is a descriptive one-instance score step, not an interval or equivalence test. Aligned probe/baseline predictions were not retained.

language	model	problems	probe F1	best baseline	difference	ρ	selectivity
Java	Qwen2.5-1.5B	1,427	0.982	0.980	+0.002	0.0120	+0.255
Java	Qwen2.5-Coder-1.5B	1,427	0.976	0.980	-0.004	0.0079	+0.271
Java	StarCoder2-7B	1,427	0.990	0.980	+0.010	0.0120	+0.243
JavaScript	Qwen2.5-1.5B	1,351	0.989	0.992	-0.003	0.0170	+0.132
JavaScript	Qwen2.5-Coder-1.5B	1,351	0.998	0.992	+0.006	0.0170	+0.150
JavaScript	StarCoder2-7B	1,351	0.988	0.992	-0.004	0.0170	+0.120
PHP	Qwen2.5-1.5B	1,305	0.983	0.969	+0.014	0.0080	+0.214
PHP	Qwen2.5-Coder-1.5B	1,305	0.975	0.969	+0.006	0.0080	+0.216
PHP	StarCoder2-7B	1,305	0.990	0.969	+0.021	0.0080	+0.240
Python	Qwen2.5-1.5B	1,301	0.982	0.983	-0.001	0.0068	+0.280
Python	Qwen2.5-Coder-1.5B	1,301	0.981	0.983	-0.002	0.0068	+0.272
Python	StarCoder2-7B	1,301	0.988	0.983	+0.005	0.0068	+0.288

Table 5: Boolean Python identifier-renaming audit. C1–C5 are the five committed renaming conditions. Each condition refits a probe and reselects its layer, so deltas measure recoverability rather than fixed-direction invariance. Intervals are problem-clustered. Empty renaming fields for other languages indicate that those runs were not committed, not a null result.

model	C1	C2	C3	C4	C5
Qwen2.5-1.5B	+0.003	-0.004	+0.001	-0.003	+0.001
Qwen2.5-Coder-1.5B	+0.001	-0.010	+0.006	-0.004	-0.006
StarCoder2-7B	-0.003	-0.013	-0.006	-0.001	-0.006

Renaming diagnostic. The misleading strategy uses role-conditioned vocabularies. Target and other identifiers receive names from different pools, so a refit probe can recover the label lexically. The all-same strategy is non-bijective and can alter label extraction. These rows diagnose intervention confounds rather than semantic robustness.

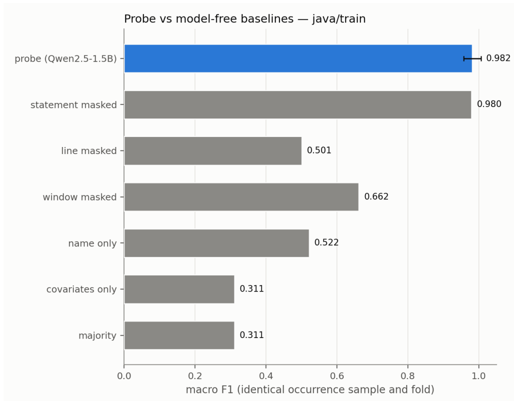
Table 6: Complete model-free boolean baselines by language. Baselines are shared across the three neural models because they use the same frozen examples and surface features.

language	majority	name	statement	line	window	covariates
Java	0.311	0.522	0.980	0.501	0.662	0.311
JavaScript	0.227	0.373	0.992	0.369	0.552	0.226
PHP	0.228	0.505	0.969	0.413	0.581	0.228
Python	0.217	0.422	0.970	0.983	0.605	0.216

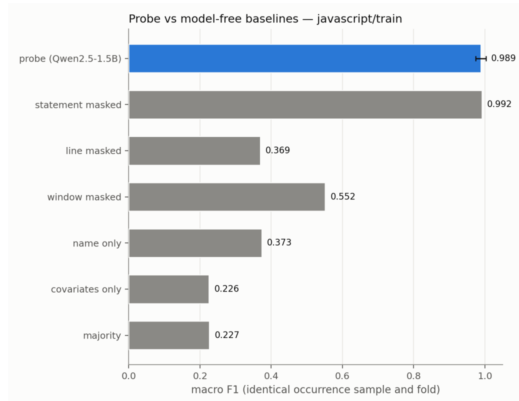
Boolean probe figures. The baseline panels retain the individual surface comparators hidden by the best-baseline table. Renaming is available only for Python. The absence of corresponding figures in other languages is a coverage hole.

Table 7: Complete class/structure perturbation results under the shared five-seed protocol. Each condition refits a probe and selects its own layer on validation data, so cross-condition deltas do not test a fixed direction.

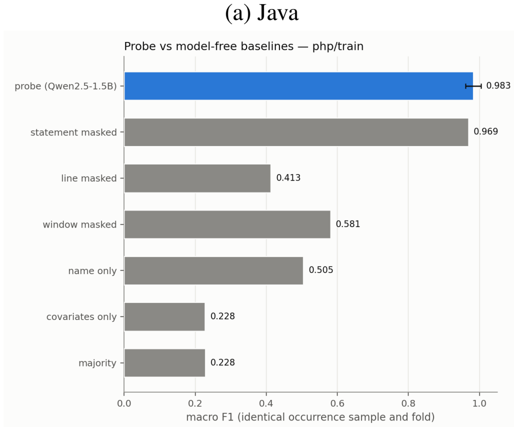
model	strategy	layer	F1	95% CI	control F1	selectivity	Δ
Qwen2.5-1.5B	baseline	L18	0.979	[0.967, 0.987]	0.427	+0.552	N/A
Qwen2.5-1.5B	random_nouns	L20	0.986	[0.976, 0.992]	0.426	+0.560	+0.007
Qwen2.5-1.5B	single_chars	L11	0.957	[0.936, 0.969]	0.423	+0.534	-0.022
Qwen2.5-1.5B	all_same	L0	1.000	[1.000, 1.000]	0.440	+0.560	+0.021
Qwen2.5-1.5B	numeric_vars	L21	0.965	[0.952, 0.972]	0.422	+0.543	-0.014
Qwen2.5-1.5B	misleading_class_struct	L7	0.976	[0.962, 0.982]	0.416	+0.559	-0.004
Qwen2.5-Coder-1.5B	baseline	L8	0.982	[0.969, 0.989]	0.429	+0.553	N/A
Qwen2.5-Coder-1.5B	random_nouns	L19	0.978	[0.965, 0.985]	0.427	+0.551	-0.004
Qwen2.5-Coder-1.5B	single_chars	L7	0.956	[0.936, 0.967]	0.423	+0.533	-0.026
Qwen2.5-Coder-1.5B	all_same	L0	1.000	[1.000, 1.000]	0.440	+0.560	+0.018
Qwen2.5-Coder-1.5B	numeric_vars	L21	0.968	[0.955, 0.975]	0.424	+0.544	-0.014
Qwen2.5-Coder-1.5B	misleading_class_struct	L7	0.978	[0.967, 0.984]	0.415	+0.562	-0.004
StarCoder2-7B	baseline	L5	0.981	[0.967, 0.990]	0.478	+0.503	N/A
StarCoder2-7B	random_nouns	L6	0.976	[0.962, 0.983]	0.466	+0.510	-0.006
StarCoder2-7B	single_chars	L4	0.957	[0.921, 0.971]	0.475	+0.482	-0.025
StarCoder2-7B	all_same	L0	1.000	[1.000, 1.000]	0.445	+0.555	+0.019
StarCoder2-7B	numeric_vars	L23	0.968	[0.956, 0.975]	0.451	+0.517	-0.013
StarCoder2-7B	misleading_class_struct	L5	0.985	[0.976, 0.990]	0.467	+0.518	+0.004



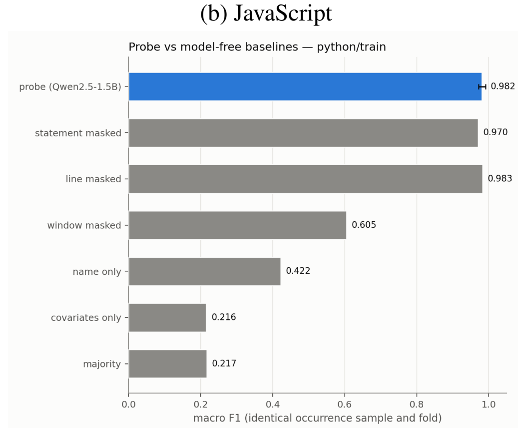
(a) Java



(b) JavaScript



(c) PHP

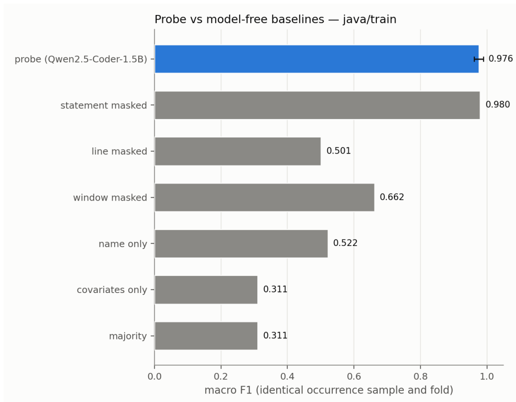


(d) Python

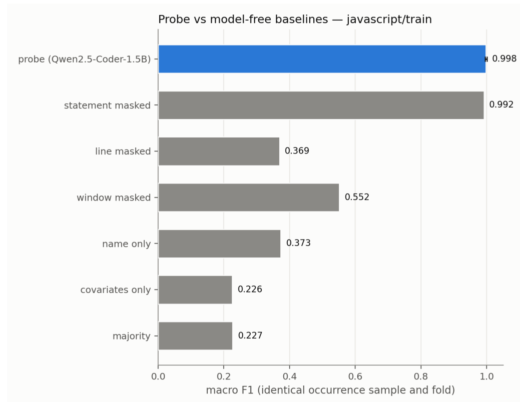
Figure 1: Boolean probe and model-free baselines for Qwen2.5-1.5B across the four committed languages.

Table 8: Class/structure cross-language results. Transfer uses the Python-selected layer without a shared source-target problem partition. In-domain layers are selected separately. Only languages present in the committed exports are shown.

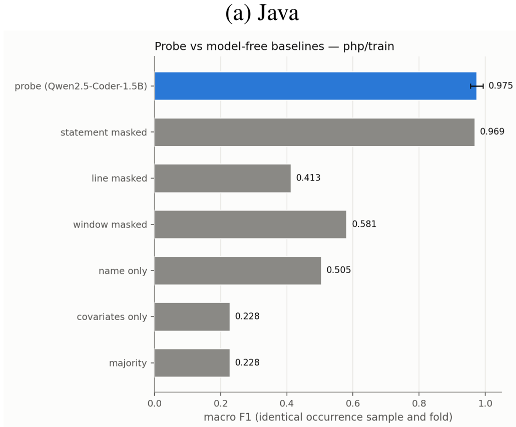
model	language	programs	in-domain layer	in-domain F1	transfer accuracy	transfer F1
Qwen2.5-1.5B	Python	400	L18	0.979	N/A	N/A
Qwen2.5-1.5B	C++	606	L10	0.991	0.993	0.933
Qwen2.5-1.5B	JavaScript	285	L5	0.987	0.998	0.957
Qwen2.5-1.5B	C	97	L8	0.986	0.984	0.886
Qwen2.5-Coder-1.5B	Python	400	L8	0.982	N/A	N/A
Qwen2.5-Coder-1.5B	C++	606	L8	0.988	0.991	0.918
Qwen2.5-Coder-1.5B	JavaScript	285	L6	0.985	0.996	0.913
Qwen2.5-Coder-1.5B	C	97	L20	0.980	0.983	0.885
StarCoder2-7B	Python	400	L5	0.981	N/A	N/A
StarCoder2-7B	C++	606	L5	0.992	0.996	0.965
StarCoder2-7B	JavaScript	285	L3	0.989	0.999	0.980
StarCoder2-7B	C	97	L5	0.986	0.987	0.906



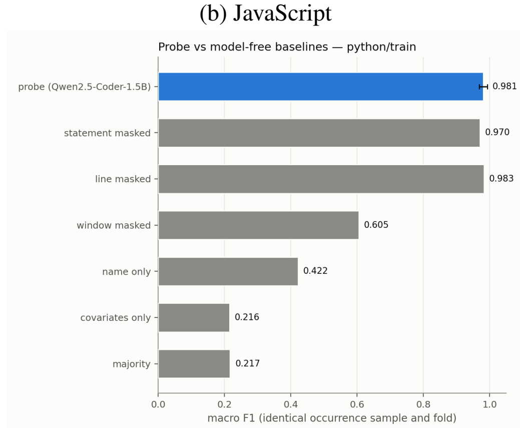
(a) Java



(b) JavaScript



(c) PHP

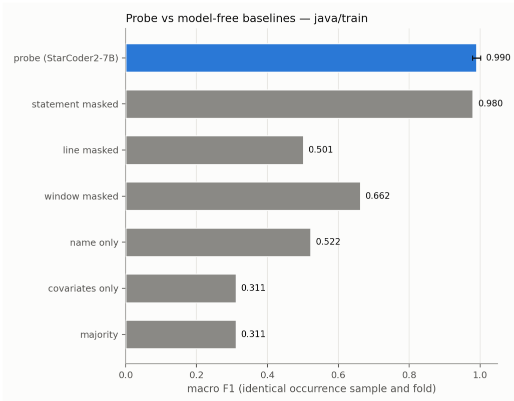


(d) Python

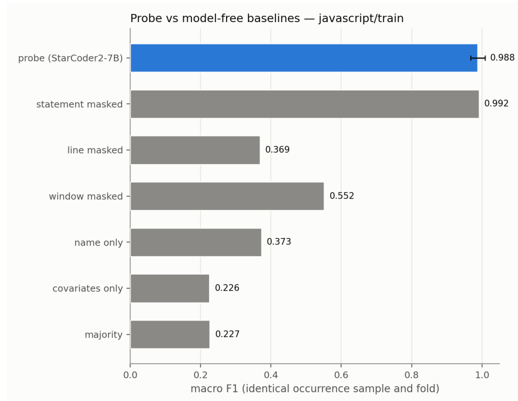
Figure 2: Boolean probe and model-free baselines for Qwen2.5-Coder-1.5B across the four committed languages.

Table 9: Layerwise class/structure robustness summaries. The F1 range is taken over all committed layers. Maximum cosine similarity is measured against the baseline activation direction for each renamed condition.

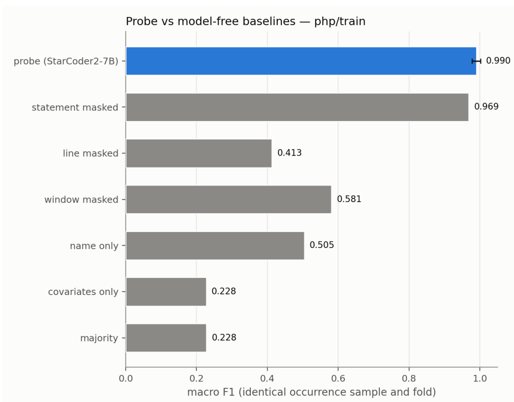
model	strategy	minimum F1	maximum F1 (layer)	maximum cosine (layer)
Qwen2.5-1.5B	all_same	0.999	1.000 (L0)	0.105 (L11)
Qwen2.5-1.5B	baseline	0.806	0.983 (L18)	N/A
Qwen2.5-1.5B	misleading_class_struct	0.708	0.980 (L7)	0.311 (L15)
Qwen2.5-1.5B	numeric_vars	0.445	0.966 (L21)	0.329 (L22)
Qwen2.5-1.5B	random_nouns	0.529	0.989 (L15)	0.493 (L26)
Qwen2.5-1.5B	single_chars	0.530	0.965 (L7)	0.392 (L15)
Qwen2.5-Coder-1.5B	all_same	0.999	1.000 (L0)	0.126 (L28)
Qwen2.5-Coder-1.5B	baseline	0.807	0.985 (L7)	N/A
Qwen2.5-Coder-1.5B	misleading_class_struct	0.708	0.978 (L7)	0.324 (L7)
Qwen2.5-Coder-1.5B	numeric_vars	0.445	0.969 (L24)	0.352 (L22)
Qwen2.5-Coder-1.5B	random_nouns	0.530	0.983 (L23)	0.464 (L19)
Qwen2.5-Coder-1.5B	single_chars	0.530	0.959 (L7)	0.415 (L15)
StarCoder2-7B	all_same	0.999	1.000 (L0)	0.101 (L21)
StarCoder2-7B	baseline	0.763	0.984 (L6)	N/A
StarCoder2-7B	misleading_class_struct	0.717	0.985 (L3)	0.387 (L3)
StarCoder2-7B	numeric_vars	0.443	0.971 (L26)	0.369 (L22)
StarCoder2-7B	random_nouns	0.502	0.979 (L8)	0.511 (L3)
StarCoder2-7B	single_chars	0.529	0.962 (L3)	0.467 (L6)



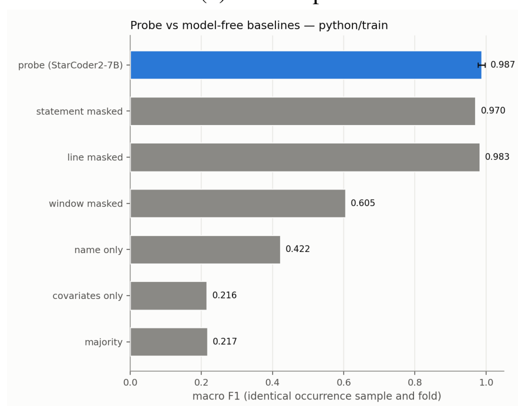
(a) Java



(b) JavaScript



(c) PHP

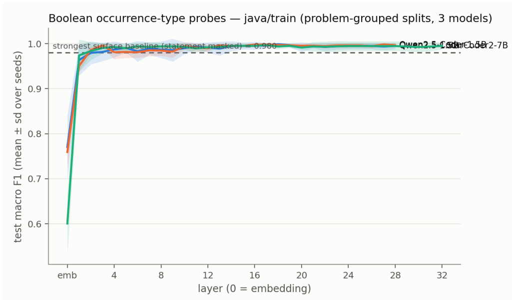


(d) Python

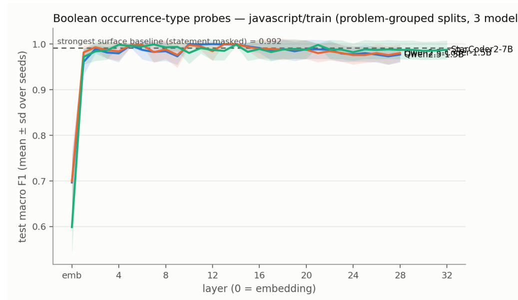
Figure 3: Boolean probe and model-free baselines for StarCoder2-7B across the four committed languages.

Table 10: Complete iterator perturbation results under the shared five-seed protocol. Each condition refits a probe and selects its own layer on validation data. Role-conditioned and non-bijective strategies are confounded diagnostics rather than robustness tests.

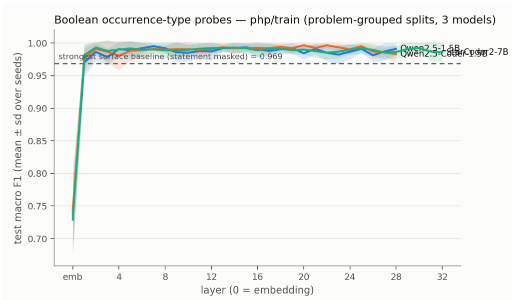
model	strategy	layer	F1	95% CI	control F1	selectivity	Δ
Qwen2.5-1.5B	baseline	L7	0.976	[0.964, 0.982]	0.424	+0.552	N/A
Qwen2.5-1.5B	random_nouns	L11	0.961	[0.951, 0.967]	0.423	+0.538	-0.015
Qwen2.5-1.5B	single_chars	L9	0.906	[0.896, 0.917]	0.428	+0.479	-0.069
Qwen2.5-1.5B	all_same	L1	1.000	[1.000, 1.000]	0.456	+0.544	+0.024
Qwen2.5-1.5B	numeric_vars	L23	0.910	[0.899, 0.920]	0.440	+0.471	-0.065
Qwen2.5-1.5B	misleading_iterator	L0	0.999	[0.998, 0.999]	0.403	+0.595	+0.023
Qwen2.5-Coder-1.5B	baseline	L7	0.978	[0.968, 0.983]	0.429	+0.549	N/A
Qwen2.5-Coder-1.5B	random_nouns	L12	0.956	[0.946, 0.962]	0.423	+0.533	-0.022
Qwen2.5-Coder-1.5B	single_chars	L8	0.906	[0.895, 0.916]	0.426	+0.480	-0.072
Qwen2.5-Coder-1.5B	all_same	L1	1.000	[1.000, 1.000]	0.459	+0.541	+0.022
Qwen2.5-Coder-1.5B	numeric_vars	L22	0.911	[0.896, 0.921]	0.440	+0.470	-0.068
Qwen2.5-Coder-1.5B	misleading_iterator	L0	0.998	[0.997, 0.999]	0.402	+0.596	+0.020
StarCoder2-7B	baseline	L11	0.987	[0.981, 0.990]	0.461	+0.526	N/A
StarCoder2-7B	random_nouns	L6	0.975	[0.967, 0.979]	0.455	+0.520	-0.012
StarCoder2-7B	single_chars	L6	0.941	[0.933, 0.949]	0.458	+0.483	-0.046
StarCoder2-7B	all_same	L2	1.000	[1.000, 1.000]	0.460	+0.540	+0.013
StarCoder2-7B	numeric_vars	L26	0.934	[0.921, 0.941]	0.454	+0.480	-0.053
StarCoder2-7B	misleading_iterator	L0	0.998	[0.997, 0.999]	0.406	+0.592	+0.012



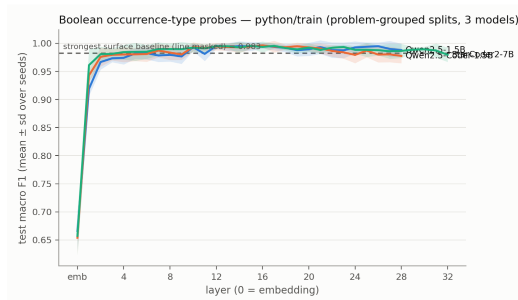
(a) Java



(b) JavaScript



(c) PHP



(d) Python

Figure 4: Boolean probe layer curves per language across the three models.

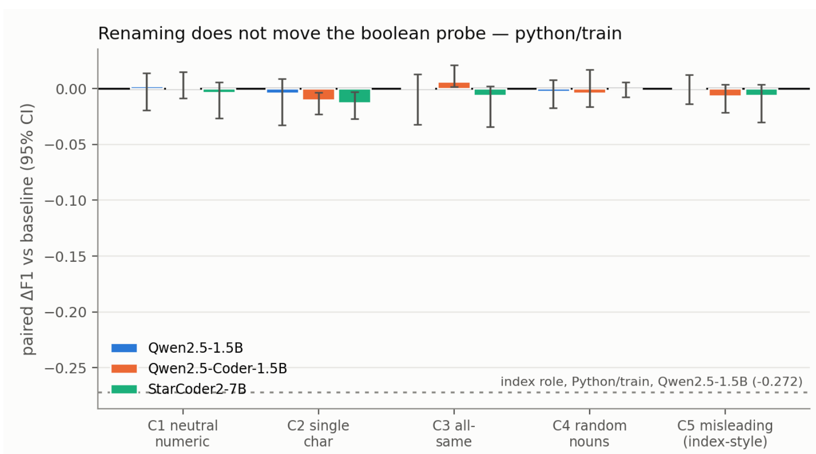


Figure 5: Problem-paired boolean refit-performance deltas on Python. No corresponding committed renaming run exists for other languages.

Table 11: Model-free iterator surface baselines on the same 427 Python programs. Scores are occurrence-level macro-F1 and are not a paired probe-minus-surface interval.

comparator	macro-F1
majority	0.363
covariates	0.361
masked line	0.844
masked statement	0.842
masked window	0.869
name only	0.918

Table 12: Complete iterator cross-language transfer results. Transfer columns use the Python-selected layer.

model	language	programs	in-domain layer	in-domain F1	transfer accuracy	transfer F1
Qwen2.5-1.5B	Python	256	L9	0.971	N/A	N/A
Qwen2.5-1.5B	Java	188	L13	0.990	0.993	0.956
Qwen2.5-1.5B	C++	191	L12	0.992	0.994	0.965
Qwen2.5-1.5B	C	191	L7	0.972	0.987	0.934
Qwen2.5-1.5B	C#	189	L13	0.987	0.993	0.959
Qwen2.5-1.5B	JavaScript	266	L9	0.985	0.992	0.966
Qwen2.5-1.5B	PHP	253	L12	0.986	0.996	0.981
Qwen2.5-Coder-1.5B	Python	256	L10	0.978	N/A	N/A
Qwen2.5-Coder-1.5B	Java	188	L11	0.989	0.986	0.908
Qwen2.5-Coder-1.5B	C++	191	L12	0.988	0.988	0.923
Qwen2.5-Coder-1.5B	C	191	L6	0.967	0.982	0.898
Qwen2.5-Coder-1.5B	C#	189	L14	0.984	0.985	0.909
Qwen2.5-Coder-1.5B	JavaScript	266	L10	0.982	0.983	0.915
Qwen2.5-Coder-1.5B	PHP	253	L12	0.988	0.986	0.923
StarCoder2-7B	Python	256	L9	0.986	N/A	N/A
StarCoder2-7B	Java	188	L7	0.992	0.980	0.887
StarCoder2-7B	C++	191	L6	0.989	0.984	0.906
StarCoder2-7B	C	191	L10	0.971	0.981	0.901
StarCoder2-7B	C#	189	L8	0.987	0.980	0.889
StarCoder2-7B	JavaScript	266	L7	0.989	0.980	0.911
StarCoder2-7B	PHP	253	L12	0.989	0.987	0.938

Table 13: Layerwise iterator robustness summaries. F1 ranges cover every committed layer. Cosine similarity compares each perturbation direction with the baseline direction.

model	strategy	minimum F1	maximum F1 (layer)	maximum cosine (layer)
Qwen2.5-1.5B	all_same	0.999	1.000 (L0)	0.200 (L15)
Qwen2.5-1.5B	baseline	0.893	0.979 (L11)	N/A
Qwen2.5-1.5B	misleading_iterator	0.988	0.999 (L8)	0.129 (L7)
Qwen2.5-1.5B	numeric_vars	0.550	0.913 (L23)	0.255 (L19)
Qwen2.5-1.5B	random_nouns	0.628	0.964 (L12)	0.392 (L19)
Qwen2.5-1.5B	single_chars	0.658	0.918 (L7)	0.265 (L21)
Qwen2.5-Coder-1.5B	all_same	0.999	1.000 (L0)	0.227 (L18)
Qwen2.5-Coder-1.5B	baseline	0.893	0.979 (L10)	N/A
Qwen2.5-Coder-1.5B	misleading_iterator	0.979	0.999 (L4)	0.179 (L20)
Qwen2.5-Coder-1.5B	numeric_vars	0.550	0.913 (L23)	0.277 (L19)
Qwen2.5-Coder-1.5B	random_nouns	0.628	0.961 (L16)	0.448 (L14)
Qwen2.5-Coder-1.5B	single_chars	0.658	0.912 (L8)	0.299 (L17)
StarCoder2-7B	all_same	0.999	1.000 (L0)	0.169 (L20)
StarCoder2-7B	baseline	0.890	0.988 (L6)	N/A
StarCoder2-7B	misleading_iterator	0.990	0.998 (L0)	0.234 (L21)
StarCoder2-7B	numeric_vars	0.549	0.937 (L24)	0.271 (L25)
StarCoder2-7B	random_nouns	0.603	0.975 (L5)	0.385 (L10)
StarCoder2-7B	single_chars	0.657	0.946 (L5)	0.359 (L12)

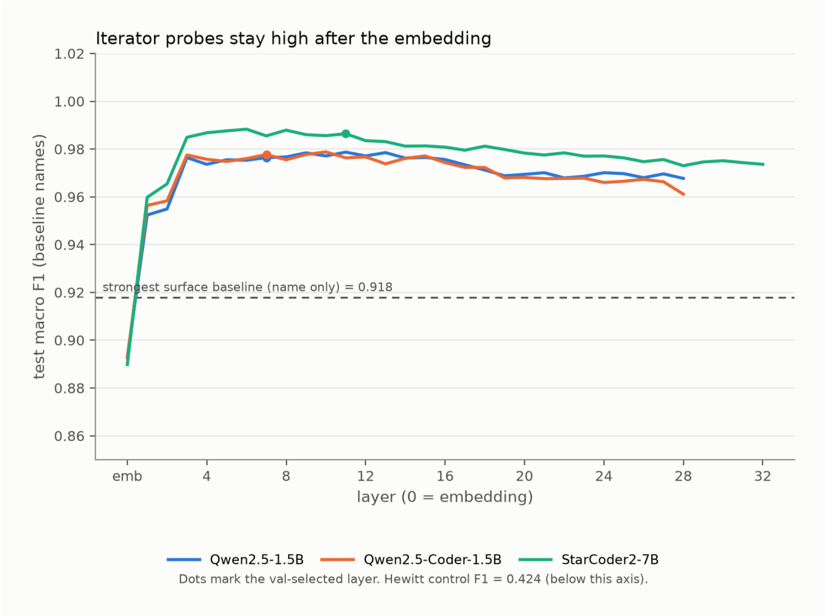


Figure 6: Iterator probe macro-F1 across layers on original names, with the name-only surface baseline marked.

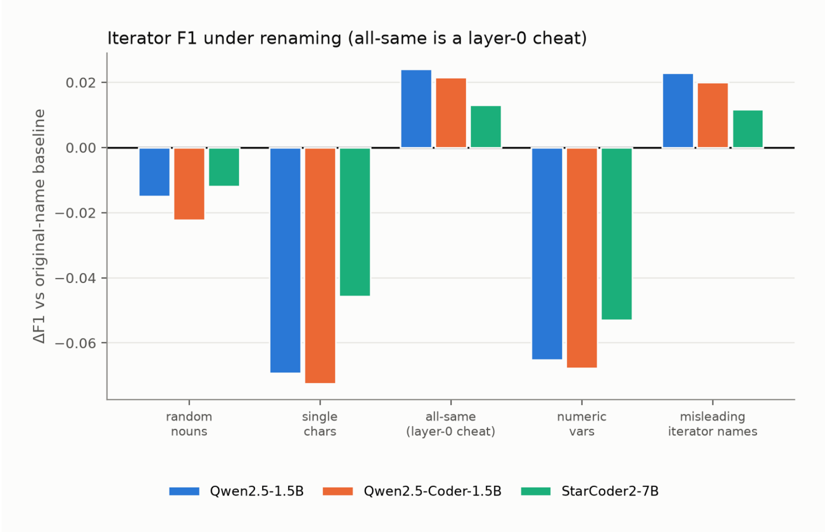


Figure 7: Iterator perturbation deltas relative to baseline.

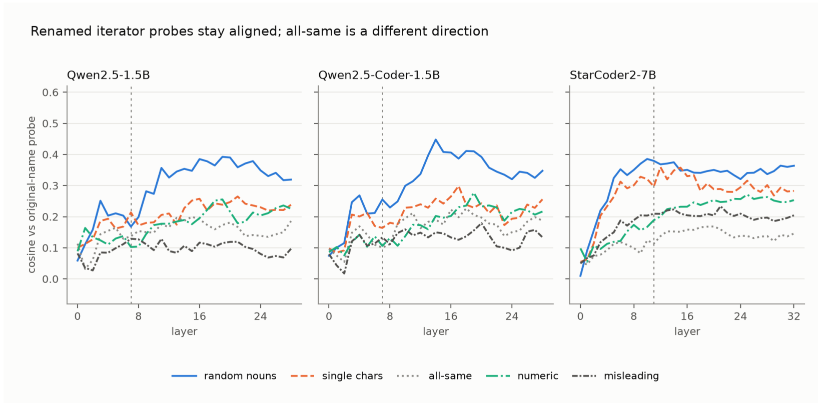


Figure 8: Iterator cosine similarity to the baseline direction.

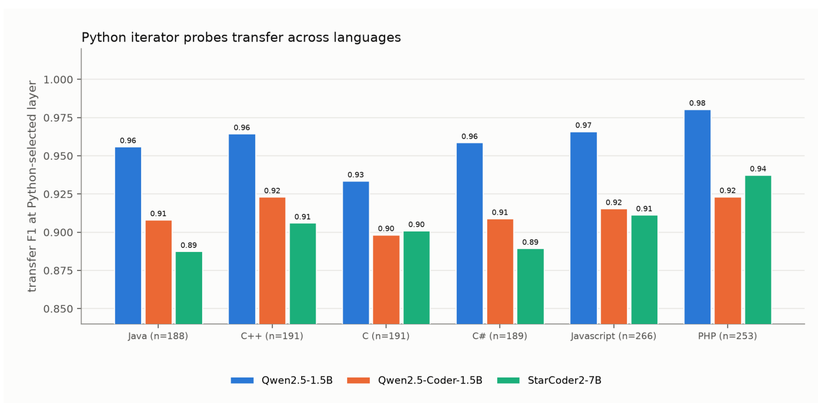


Figure 9: Available iterator cross-language transfer results.

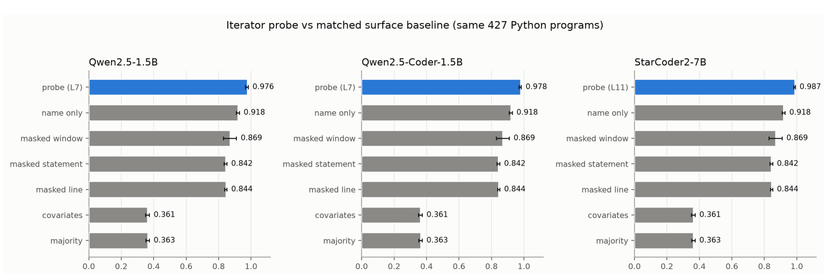


Figure 10: Iterator probe versus matched model-free surface baselines on the same 427 Python programs.

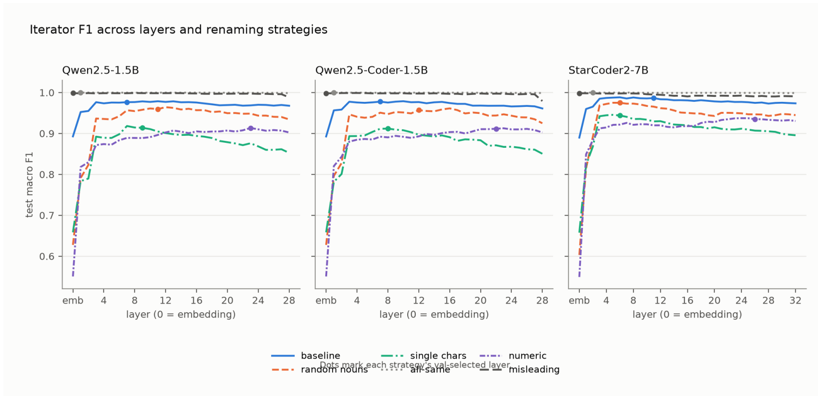


Figure 11: Iterator probe macro-F1 across layers and renaming strategies.

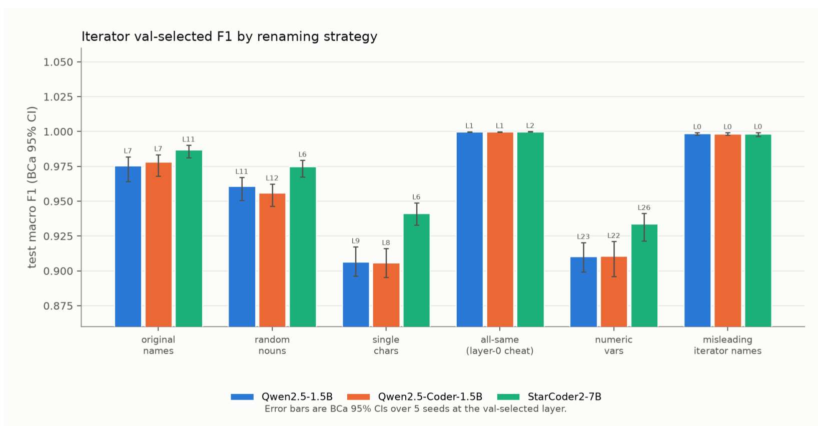


Figure 12: Iterator val-selected F1 by renaming strategy, with BCa 95% intervals.

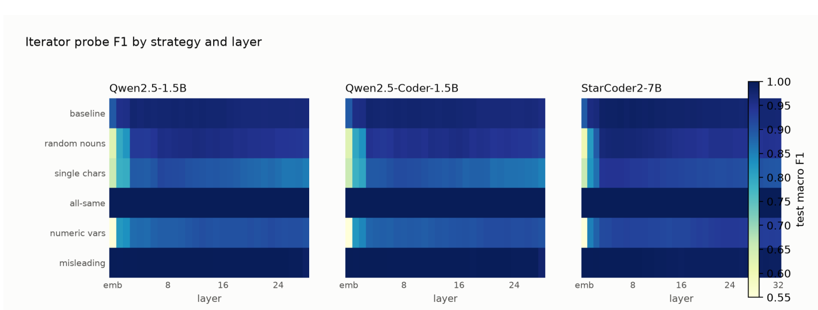


Figure 13: Iterator probe F1 by strategy and layer.

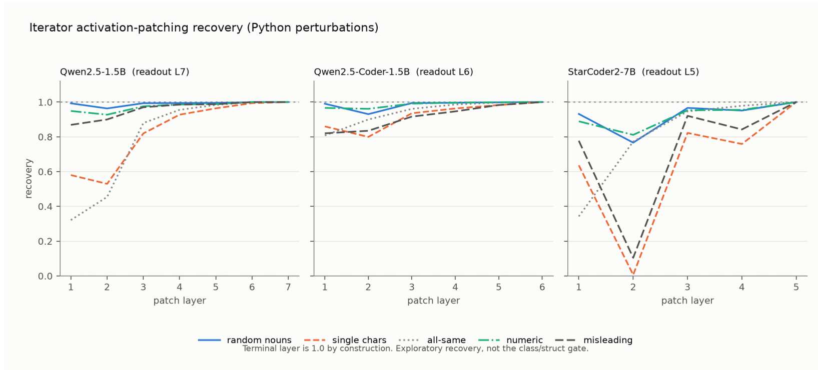


Figure 14: Exploratory iterator activation-patching recovery under Python name perturbations. The terminal layer is one by construction.

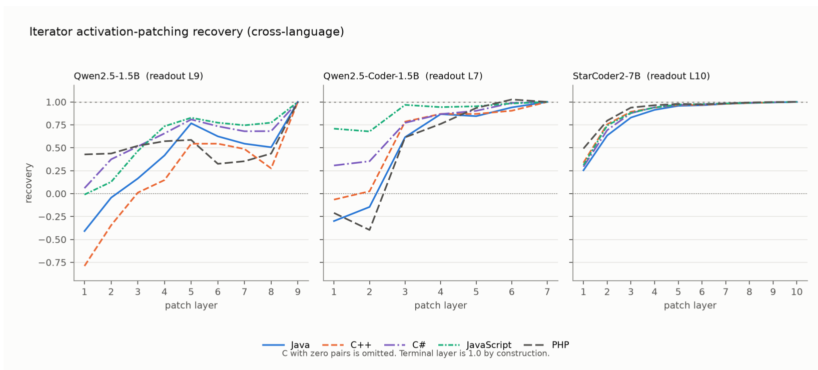


Figure 15: Exploratory iterator activation-patching recovery under cross-language transfer. C-language rows with zero pairs are omitted.

B Causal intervention details

Table 14: Complete gate-specified class/structure patching summary in Qwen2.5-1.5B. The causal gate required mean effect at least 0.10, a positive interval, separation from controls, and recovery at least 0.05 in both directions.

layer	span	direction	control	n	mean effect	95% CI	recovery	minus random
0	query_name	denoise	target	288	+0.0000	[0.0000, 0.0000]	+0.0000	N/A
0	query_name	noise	target	288	+0.0000	[0.0000, 0.0000]	+0.0000	N/A
18	placebo	denoise	target	288	+0.0015	[-0.0010, 0.0043]	+0.0015	N/A
18	placebo	noise	target	288	+0.0006	[-0.0019, 0.0032]	+0.0006	N/A
18	query_name	denoise	random	288	-0.0107	[-0.0205, -0.0014]	-0.0110	N/A
18	query_name	denoise	same_source	288	+0.0004	[-0.0009, 0.0016]	+0.0004	N/A
18	query_name	denoise	target	288	+0.0087	[-0.0007, 0.0188]	+0.0089	+0.0194
18	query_name	noise	random	288	+0.0096	[0.0023, 0.0164]	+0.0099	N/A
18	query_name	noise	same_source	288	-0.0005	[-0.0018, 0.0006]	-0.0006	N/A
18	query_name	noise	target	288	+0.0199	[0.0109, 0.0296]	+0.0204	+0.0103

Status. The iterator recovery exports predate the gate-specified class/structure protocol. They report descriptive recovery curves without clustered confidence intervals or matched placebo/random controls. The terminal value is one by construction, and C-language cross-language rows with zero pairs are missing data rather than null effects.

Table 15: Aggregate probe-link diagnostic for the class/structure evaluation pairs. Absolute values are reported because pair orientation is arbitrary. Large probe movement alongside small behavioral movement is descriptive of the failed causal transfer, not evidence that probe movement causes behavior.

quantity	mean absolute	median absolute
probe movement	15.786	16.302
behavioral movement	0.038	0.031

Table 16: Class/structure gate and readout audit. Both prompt classes favor `True`. The pairwise gap remains separable, but the binary readout is not calibrated to distinguish function prompts.

diagnostic	estimate	95% CI	gate
class logit difference	2.752	[2.624, 2.898]	pass
function logit difference	1.778	[1.652, 1.909]	fail
class-minus-function gap	0.974	[0.908, 1.035]	pass
pair-gap accuracy	0.986	[0.962, 0.993]	pass
query/declaration probe AUC	0.943/0.905	N/A	pass

Table 17: Patching audit trail. Completeness establishes that the scheduled Qwen evaluation finished. It does not turn the failed causal gate into positive evidence. The controller stopped before Coder and StarCoder2 after the Qwen null.

audit item	value
behavior rows present/expected	1152/1152
primary rows present/expected	4032/4032
baseline probe-gap/behavior Spearman	0.164 [0.023, 0.286]
controller terminal state	`stopped_qwen_null`

Table 18: Exploratory iterator activation-patching recovery for Qwen2.5-1.5B, within-language perturbations.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5	L6	L7
random_nouns	L7	150	0.999/0.307	0.993	0.963	0.994	0.994	0.995	0.999	1.000
single_chars	L7	150	0.995/0.712	0.580	0.530	0.820	0.927	0.964	0.994	1.000
all_same	L7	150	0.999/0.138	0.321	0.454	0.879	0.955	0.984	1.000	1.000
numeric_vars	L7	150	0.999/0.211	0.949	0.927	0.976	0.986	0.989	1.000	1.000
misleading_iterator	L7	150	0.998/0.324	0.869	0.900	0.970	0.985	0.988	1.000	1.000

Table 19: Exploratory iterator activation-patching recovery for Qwen2.5-1.5B, cross-language transfer.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5	L6	L7	L8	L9
Java	L9	20	0.998/0.768	-0.409	-0.044	0.165	0.416	0.767	0.625	0.544	0.505	1.000
C++	L9	20	0.996/0.801	-0.790	-0.345	0.010	0.148	0.544	0.545	0.486	0.275	1.000
C	L9	0	0.000/0.000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
C#	L9	30	0.993/0.801	0.059	0.374	0.514	0.657	0.809	0.734	0.680	0.681	1.000
JavaScript	L9	44	0.996/0.828	-0.011	0.128	0.465	0.734	0.827	0.773	0.745	0.774	1.000
PHP	L9	12	0.987/0.810	0.427	0.437	0.521	0.571	0.586	0.325	0.353	0.437	1.000

Table 20: Exploratory iterator activation-patching recovery for Qwen2.5-Coder-1.5B, within-language perturbations.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5	L6
random_nouns	L6	150	0.999/0.138	0.991	0.930	0.994	0.996	0.998	1.000
single_chars	L6	150	0.996/0.614	0.860	0.799	0.935	0.963	0.983	1.000
all_same	L6	150	0.999/0.142	0.808	0.900	0.961	0.986	0.998	1.000
numeric_vars	L6	150	0.999/0.057	0.966	0.961	0.991	0.997	0.999	1.000
misleading_iterator	L6	150	0.999/0.286	0.820	0.835	0.916	0.946	0.983	1.000

Table 21: Exploratory iterator activation-patching recovery for Qwen2.5-Coder-1.5B, cross-language transfer.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5	L6	L7
Java	L7	10	0.989/0.709	-0.300	-0.146	0.616	0.867	0.845	0.940	1.000
C++	L7	11	0.979/0.769	-0.066	0.025	0.785	0.869	0.869	0.904	1.000
C	L7	0	0.000/0.000	N/A	N/A	N/A	N/A	N/A	N/A	N/A
C#	L7	9	0.981/0.629	0.307	0.353	0.772	0.867	0.903	0.989	1.000
JavaScript	L7	13	0.983/0.766	0.708	0.678	0.968	0.943	0.953	0.983	1.000
PHP	L7	7	0.982/0.748	-0.211	-0.397	0.614	0.760	0.934	1.027	1.000

Table 22: Exploratory iterator activation-patching recovery for StarCoder2-7B, within-language perturbations.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5
random_nouns	L5	150	0.998/0.569	0.931	0.766	0.966	0.951	1.000
single_chars	L5	150	0.994/0.686	0.636	0.006	0.823	0.759	1.000
all_same	L5	150	0.999/0.249	0.342	0.772	0.946	0.979	1.000
numeric_vars	L5	150	0.998/0.530	0.889	0.811	0.953	0.955	1.000
misleading_iterator	L5	150	0.997/0.628	0.777	0.104	0.921	0.843	1.000

Table 23: Exploratory iterator activation-patching recovery for StarCoder2-7B, cross-language transfer.

condition	readout	pairs	clean/corrupt	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
Java	L10	150	0.996/0.750	0.254	0.634	0.829	0.913	0.957	0.964	0.979	0.988	0.994	1.000
C++	L10	150	0.996/0.771	0.337	0.756	0.892	0.938	0.966	0.969	0.979	0.988	0.995	1.000
C	L10	0	0.000/0.000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
C#	L10	150	0.996/0.740	0.294	0.692	0.877	0.942	0.969	0.974	0.984	0.991	0.996	1.000
JavaScript	L10	150	0.996/0.716	0.311	0.743	0.880	0.939	0.972	0.976	0.983	0.990	0.996	1.000
PHP	L10	73	0.993/0.808	0.491	0.796	0.938	0.964	0.979	0.973	0.983	0.994	0.999	1.000

C Exploratory boolean interventions

Status and estimand. These boolean interventions are exploratory diagnostics rather than evidence for the main causal claim. The committed export contains layer-wise means but not the per-case arrays needed to recompute clustered intervals, and the runs do not use the specified gate of the class/structure experiment. The exporter defines effect as $|\mathrm{clean} - \mathrm{intervened}|$ and specificity as that effect divided by the corresponding random-position-control effect. Specificity can become large when its denominator is small. We therefore report both specificity at the largest-effect layer and the maximum specificity over layers rather than selecting only the more favorable one.

Table 24: Exploratory boolean patch diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

language	model	n	peak layer	peak effect	spec. at peak	best spec. (layer)
C++	Qwen2.5-1.5B	157	L0	14.845	3.6×	7.9× (L20)
C++	Qwen2.5-Coder-1.5B	158	L0	15.414	3.6×	6.9× (L8)
C++	StarCoder2-7B	157	L4	14.982	8.1×	10.4× (L16)
Java	Qwen2.5-1.5B	155	L8	11.607	6.1×	7.7× (L12)
Java	Qwen2.5-Coder-1.5B	155	L8	12.642	5.9×	7.7× (L12)
Java	StarCoder2-7B	155	L4	13.900	7.5×	8.6× (L12)
JavaScript	Qwen2.5-1.5B	58	L8	12.755	6.9×	6.9× (L8)
JavaScript	Qwen2.5-Coder-1.5B	62	L8	12.717	6.6×	7.4× (L12)
JavaScript	StarCoder2-7B	129	L4	12.137	8.1×	8.2× (L16)
PHP	Qwen2.5-1.5B	15	L4	11.698	7.2×	7.2× (L4)
PHP	Qwen2.5-Coder-1.5B	16	L4	11.425	4.7×	7.4× (L0)
PHP	StarCoder2-7B	17	L4	11.545	6.3×	9.3× (L16)
Python	Qwen2.5-1.5B	176	L8	12.948	6.0×	7.2× (L16)
Python	Qwen2.5-Coder-1.5B	176	L8	13.275	6.7×	7.8× (L16)
Python	StarCoder2-7B	176	L4	13.592	7.9×	9.7× (L12)

Table 25: Exploratory boolean steer diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

language	model	n	peak layer	peak effect	spec. at peak	best spec. (layer)
C++	Qwen2.5-1.5B	79	L24	2.508	45.0×	169.3× (L4)
C++	Qwen2.5-Coder-1.5B	79	L24	1.436	602.0×	602.0× (L24)
C++	StarCoder2-7B	78	L20	0.653	6.9×	345.4× (L4)
Java	Qwen2.5-1.5B	78	L24	2.486	72.2×	826.9× (L8)
Java	Qwen2.5-Coder-1.5B	78	L24	1.712	72.0×	179.9× (L0)
Java	StarCoder2-7B	78	L24	0.555	244.0×	244.0× (L24)
JavaScript	Qwen2.5-1.5B	31	L24	1.763	24.5×	76.8× (L8)
JavaScript	Qwen2.5-Coder-1.5B	31	L24	1.147	65.3×	147.6× (L4)
JavaScript	StarCoder2-7B	65	L16	0.591	11.2×	61.0× (L28)
PHP	Qwen2.5-1.5B	7	L24	2.759	190.2×	190.2× (L24)
PHP	Qwen2.5-Coder-1.5B	8	L24	0.898	11.5×	132.7× (L8)
PHP	StarCoder2-7B	9	L20	0.991	8.7×	99.9× (L4)
Python	Qwen2.5-1.5B	88	L24	2.175	209.1×	794.6× (L16)
Python	Qwen2.5-Coder-1.5B	88	L24	1.231	1027.1×	1027.1× (L24)
Python	StarCoder2-7B	88	L20	0.721	8.6×	55.1× (L4)

Omitted figures. Per-cell diagnostic plots for the exploratory boolean interventions and the committed role analyses are reproducible from the same artifacts with `make_interp_appendix.py -full-figures`. They visualize the tables above and add no uncertainty estimate the exports lack.

Table 26: Exploratory boolean ablate diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

language	model	n	peak layer	peak effect	spec. at peak	best spec. (layer)
C++	Qwen2.5-1.5B	157	L0	14.286	4.5×	31.9× (L4)
C++	Qwen2.5-Coder-1.5B	158	L0	12.927	4.4×	76.2× (L20)
C++	StarCoder2-7B	157	L0	13.405	4.3×	36.5× (L4)
Java	Qwen2.5-1.5B	155	L16	9.720	28.5×	86.0× (L12)
Java	Qwen2.5-Coder-1.5B	155	L4	9.331	12.3×	62.4× (L12)
Java	StarCoder2-7B	155	L4	11.190	27.0×	27.0× (L4)
JavaScript	Qwen2.5-1.5B	58	L16	10.852	26.7×	26.7× (L16)
JavaScript	Qwen2.5-Coder-1.5B	62	L16	9.868	28.1×	116.4× (L24)
JavaScript	StarCoder2-7B	129	L4	10.970	27.3×	27.3× (L4)
PHP	Qwen2.5-1.5B	15	L16	15.698	15.0×	102.4× (L20)
PHP	Qwen2.5-Coder-1.5B	16	L16	13.402	26.5×	79.2× (L20)
PHP	StarCoder2-7B	17	L4	14.260	19.4×	19.7× (L8)
Python	Qwen2.5-1.5B	176	L0	12.144	5.8×	36.7× (L24)
Python	Qwen2.5-Coder-1.5B	176	L0	11.286	6.8×	33.1× (L16)
Python	StarCoder2-7B	176	L4	11.818	26.5×	26.5× (L4)

D Explicit coverage holes

- Accumulator and index/key results use legacy exports. No committed problem-grouped, five-seed CSV reproduces those roles under the modern protocol.
- Iterator has a model-free surface comparator. The probe-minus-surface gap is not a paired clustered interval.
- Class/structure transfer exports include Python, C++, JavaScript, and C, but not Java, C#, or PHP.
- Boolean probe exports include Python, Java, JavaScript, and PHP. C, C#, and C++ are absent, and renaming is available only for Python.
- Gate-specified class/structure patching stopped after the Qwen2.5-1.5B null. Coder, StarCoder2, and the all-layer core sweep were not attempted.
- No matched model-free surface comparator is committed for the class/structure role.

E Reproducibility

The appendix is generated from committed aggregate boolean, iterator, class or structure, and patching artifacts with `uv run python scripts/make_interp_appendix.py`. The claim audit uses `uv run python tests/test_interp4d_claims.py`. Modern probes use problem hashes, validation-selected layers, and five seeds where stated. The class or structure intervention records model and dataset revisions, hashes, all $4{,}032$ rows across its primary and behavior schedules, and clustered intervals. The paired boolean CSV is aggregate evidence because its aligned prediction inputs were not retained. The regression test checks manuscript-to-artifact consistency rather than reproducing activation extraction or probe fitting.

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